



The University of Lahore

Department of CS & IT

Assignment 03

Instructor Name: Miss Iqra Adnan

Course Code: CS09202 | 11

Course Title: Discrete Structures

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Due Date: 22-12-2024

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SAP ID: 70150006

Question 1: Consider the following relations on $\{1,2,3,4\}$:

Answers:

$R1 = \{(1,1), (1,2), (2,1), (2,2), (3,4), (4,1), (4,4)\}$, **Not reflexive**

$R2 = \{(1,1), (1,2), (2,1)\}$, **Not reflexive**

$R3 = \{(1,1), (1,2), (1,4), (2,1), (2,2), (3,3), (4,1), (4,4)\}$, **Reflexive**

$R4 = \{(2,1), (3,1), (3,2), (4,1), (4,2), (4,3)\}$, **Not reflexive**

$R5 = \{(1,1), (1,2), (1,3), (1,4), (2,2), (2,3), (2,4), (3,3), (3,4), (4,4)\}$, **Reflexive**

$R6 = \{(3,4)\}$. **Not reflexive**

Which of these relations are reflexive?

Question 2: Let $A = \{1,2,3\}$ and $B = \{1,2,3,4\}$.

The relations are $R1 = \{(1,1), (2,2), (3,3)\}$ and $R2 = \{(1,1), (1,2), (1,3), (1,4)\}$ combine them and obtain the following :

Answers:

$R1 \cup R2$, $= \{(1,1), (2,2), (3,3), (1,2), (1,3), (1,4)\}$

$R1 \cap R2$, $= \{(1,1)\}$

$R1 - R2$, $= \{(2,2), (3,3)\}$

$R2 - R1$. $= \{(1,2), (1,3), (1,4)\}$

Question 3: Find the in-degree and out-degree of each vertex in the graph G with directed edges shown in Figure given below .

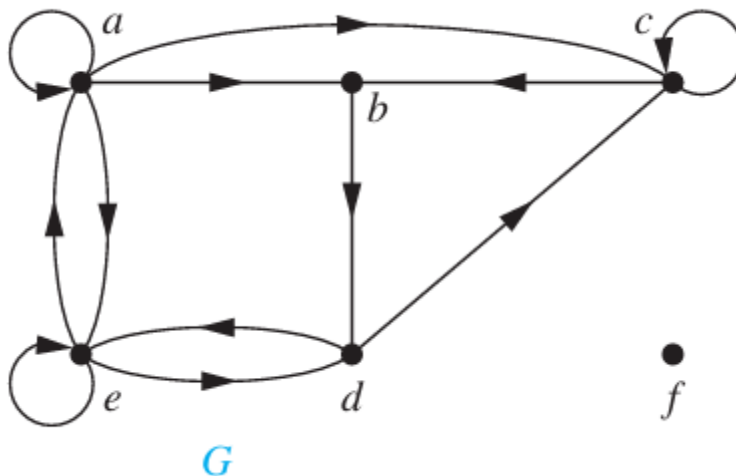
Answers:

A: in-degree = 2, out-degree = 1 (At vertex A)

B: in-degree = 2, out-degree = 2 (At vertex B)

C: in-degree = 1, out-degree = 2 (At vertex C)

D: in-degree = 1, out-degree = 1 (At vertex D)



Question 4: Show that preorder traversals of the two ordered rooted trees displayed below produce the same list of vertices.

Answers:

Tree 1 preorder traversal: a, b, d, c, e

Tree 2 preorder traversal: a, b, d, c, e

